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HUNNY GUPTA
9999903256, 9999603256

# HT Board Booster

## Class X Science

# Chemistry

## CBSE Board Paper Question Bank

Chapter Wise Questions

Ch - 1
Chemical Reactions and Equations

Session 2026-27



**Class X Science Chemistry**
**Ch- Chemical Reactions and Equations**
**Previous Year CBSE Board Question Paper**

**CBSE Board Paper 2025-26**
**Section-A (1 Marker Questions)**

Q.1. (i) $AgNO_3 + NaCl \rightarrow NaNO_3 + AgCl$ **(Set- 31/1/1)**
(ii) $K_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2KCl$
Which of the following options clearly describes both the reactions?
(A) (i) is double displacement, (ii) is displacement reaction.
(B) Both, (i) and (ii) are displacement reactions and precipitation reactions.
(C) Both, (i) and (ii) are double displacement reactions and precipitation reactions.
(D) (i) is displacement, (ii) is double displacement reaction.

Q.2. The gases evolved on heating lead (II) nitrate crystals are: **(Set- 31/1/1)**
(a) NO and $O_2$ (b) $N_2$ and $NO_2$
(c) $NO_2$ and $H_2$ (d) $NO_2$ and $O_2$

Q.3. Which of the following are combination reactions? **(Set- 31/2/1)**
(i) $2KClO_3 \rightarrow 2KCl + 3O_2$
(ii) $MgO + H_2O \rightarrow Mg(OH)_2$
(iii) $4Al + 3O_2 \rightarrow 2Al_2O_3$
(iv) $Zn + FeSO_4 \rightarrow ZnSO_4 + Fe$
Options:
(A) (i) and (iii) (B) (iii) and (iv)
(C) (ii) and (iv) (D) (ii) and (iii)

Q.4. When ferrous sulphate crystals are heated in a test tube, we observe that : **(Set- 31/2/1)**
(a) A colourless gas with no smell is evolved.
(b) A brown gas is evolved.
(c) The green colour of the salt fades and a gas with the smell of burning sulphur is evolved.
(d) The green colour of the salt fades and no gas is evolved.

Q.5. Which of the following is an example of a displacement reaction ? **(Set- 31/3/1)**
(a) The combustion of natural gas
(b) The reaction of a reactive metal with dilute acid
(c) The electrolysis of water
(d) The reaction of quick lime with water

Q.6. In the following equation : x KI + $H_2O_2 \rightarrow I_2$ + y KOH x and y respectively are: **(Set- 31/3/1)**
(a) 2, 2 (b) 1, 2 (c) 2, 1 (d) 1, 1

Q.7. The volume ratio of hydrogen and oxygen gases liberated during electrolysis of water is
(a) 1 : 1 (b) 2 : 1 (c) 4 : 1 (d) 1 : 2 **(Set- 31/4/1 & 31/5/1)**

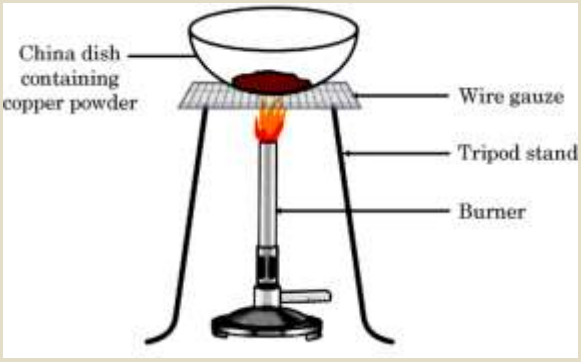
Q.8. **Assertion (A)**: Reaction of quick lime with water is an exothermic reaction. **(Set- 31/5/1)**
**Reason (R)**: A large amount of heat is evolved on the reaction of quick lime and water.

**Section-C (3 Marker Questions)**

Q.9. (a) Name the substance oxidised and reduced in the following reaction: **(Set- 31/1/1)**
$ZnO + C \rightarrow Zn + CO$
(b) Balance the following chemical reaction:
$Pb(NO_3)_2 + KI \rightarrow PbI_2 + KNO_3$
(c) Give one example each of electrolytic decomposition and decomposition by sunlight.

Hunny Gupta 9999903256

Q.10.




Seema took a small amount of copper powder in a china dish and heated it. What changes will she observe on heating? When H2 gas is passed over this heated substance, what visible changes will be seen in it? Give the chemical equations of the reactions involved, along with the names and colours of the products in each case. (**Set- 31/2/1**)

Q.11. 2 g of green coloured crystals of ferrous sulphate are heated in a dry boiling tube. Name the type of chemical reaction taking place. Write the balanced chemical equation for the reaction. Is this an exothermic or an endothermic reaction? (**Set- 31/3/1**)

Q.12. Account for the following: (**Set- 31/4/1**)

(a) White silver chloride turns grey when exposed to sunlight.
(b) Lead (II) nitrate on heating releases brown fumes.
(c) Bags of oil and fat containing food items are flushed with nitrogen gas.

Q.13. What happens when (**Set- 31/5/1**)

(i) lead (II) nitrate is heated?
(ii) iron (II) sulphate is heated?
(iii) silver chloride is exposed to sunlight?

Write balanced chemical equations of the reactions involved in support of your answer.

Hunny Gupta 9999903256

**CBSE BP 2024-25**

**Section-A (one Marker Questions)**

Q.14. Electrolysis of water is a decomposition reaction. The mass ratio ($M_H : M_O$) of hydrogen and oxygen gases liberated at the electrodes during electrolysis of water is **(Set- 31/1/1)**

(a) 8 : 1 (b) 2 : 1 (c) 1 : 2 (d) 1 : 8

Q.15. **Assertion (A):** Decomposition reactions are generally endothermic reactions. **(Set- 31/1/1)**

**Reason (R):** Decomposition of organic matter into compost is an exothermic process.

Q.16. The correct balanced chemical equation showing exothermic reaction in which natural gas burns in air is: **(Set- 31/2/1)**

Q.17. (a) $CH_4 + O_2 \rightarrow CO_2 + 2H_2O$ (b) $CH_4 + 2O_2 \rightarrow 2CO_2 + 2H_2O$ + Energy

(c) $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ (d) $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + Energy

Q.18. Consider the following chemical equation: **(Set- 31/3/1)**

$p\ Al + q\ H_2O \rightarrow r\ Al_2O_3 + s\ H_2$

To balance this chemical equation, the values of 'p', 'q', 'r' and 's' must be respectively:

(a) 3, 2, 2, 1 (b) 2, 3, 3, 1 (c) 2, 3, 1, 3 (d) 3, 1, 2, 2

Q.19. In which one of the following situations a chemical reaction does not occur? **(Set- 31/4/1)**

(a) Milk is left open at room temperature during summer

(b) Grapes get fermented

(c) An iron nail is left exposed to humid atmosphere

(d) Melting of glaciers

Q.20. The main observations while performing the experiment of burning magnesium ribbon in air are

(i) Magnesium ribbon burns with a dazzling white flame. **(Set- 31/5/1)**

(ii) A white powder is formed.

(iii) Magnesium ribbon vapourises.

(iv) Aqueous solution of the white powder turns blue litmus to red.

(a) (i) and (iv) (b) (ii) & (iii) (c) (i) and (ii) (d) (iii) & (iv)

Q.21. The values of a, b, c and d in the following balanced chemical equation are respectively

$a\ Pb(NO_3)_2 \xrightarrow{heat} b\ PbO + e\ NO_2 + d\ O_2$ **(Set- 31/4/1)**

(a) 1, 1, 2, 1 (b) 1, 1, 1, 2 (c) 2, 2, 1, 4 (d) 2, 2, 4, 1

Q.22. Assertion (A): Silver chloride turns grey in sunlight. **(Set- 31/5/1)**

Reason (R) : Decomposition of silver chloride into silver and chlorine takes place by sunlight.

Q.23. In the following experimental setup of electrolysis of water, if P and Q are the gases collected in the test tubes enclosing the electrodes R and S, then select the option/options in which the matching is correct: **(Set- 31/6/1)**

(i) P — Oxygen gas , R – Anode

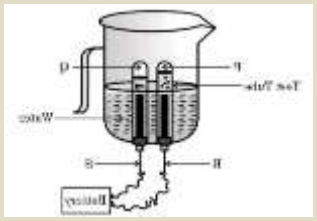
(ii) P — Hydrogen gas , R – Cathode

(iii) Q — Hydrogen gas , S - Cathode

(iv) Q - Oxygen gas , S – Anode

(a) (i) and (ii) (b) (iii) and (iv)

(b) (i) and (iii) (d) (ii) and (iv)



Q.24. Select from the following a statement which is not true about burning of magnesium ribbon in air

(a) It burns with a dazzling white flame. **(Set- 31/6/1)**

(b) A white powder is formed on burning.

(c) It is an endothermic reaction.

(d) It is an example of a combination reaction.

**Section-B**

Q.25. (a) List the possible sources of energy required in decomposition reactions. Illustrate any one with a suitable example. **(Set- 31/3/1)**

OR



Hunny Gupta 9999903256

(a) What is observed when hydrated ferrous sulphate crystals are heated in a dry boiling tube? Give balanced chemical equation(s) of the reactions(s) that occur(s).

Q.26. Define oxidation. Identify and name the substance oxidised in the following reaction :
$CuO + H_2 \rightarrow Cu + H_2O$ **(Set- 31/4/1)**

Q.27. A crystalline substance of green colour 'X' emits gases of characteristic odour when heated over a flame. It first loses water and changes colour. On further heating, it decomposes and produces a solid compound Y. **(Set- 31/5/1)**

(a) Identify 'X' and 'Y.

(b) State the change in colour observed when 'X' is heated.

**Section–C**

Q.28. A student performs the following experiment in his school laboratory. List two observations to justify that in this experiment a chemical change has taken place. **(Set- 31/1/1)**




Q.29. (i) Define the term decomposition reaction. Write one chemical equation each for decomposition reaction where energy is supplied in the form of heat, light or electricity. **(Set- 31/2/1)**

(ii) Decomposition of vegetable matter into compost is considered an exothermic reaction. Why?

**OR**

Why are decomposition reactions called the opposite of combination reactions? Write one chemical equation each for these two types of reactions mentioning the name of the reactant(s) and the product(s) involved in the reactions.

Q.30. (i) Why do we balance a chemical equation? Name and state the law that suggests the balancing of a chemical equation? Balance the following chemical equation :
$Zn + H_2PO_4 \rightarrow Zn_3(PO_4)_2 + H_2$ **(Set- 31/4/1)**

OR

(ii) Define a precipitation reaction. Give its example and also express the reaction that occurs in the form of a balanced chemical equation.

Q.31. Write balanced chemical equation for the reactions that occur when **(Set- 31/6/1)**

(a) steam is passed over red hot iron.

(b) natural gas is burnt in air.

(c) glucose reacts with oxygen in the cells of our body and provides energy.

Hunny Gupta 9999033256

**CBSE BP 2023-24**

**Section-A (one Marker Questions)**

Q.32. Select from the following a decomposition reaction in which source of energy for decomposition is light: **(Set- 31/1/1)**

(a) $2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$ (b) $2H_2O \rightarrow 2H_2 + O_2$

(c) $2AgBr \rightarrow 2Ag + Br_2$ (d) $CaCO_3 \rightarrow CaO + CO_2$

Q.33. $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$ **(Set- 31/1/1)**

The reaction given above is a redox reaction because in this case:

(a) $MnO_2$ is oxidised and HCl is reduced. (b) HCl is oxidised.

(c) $MnO_2$ is reduced. (d) $MnO_2$ is reduced and HCI is oxidised.

Q.34. Select from the following a process in which a combination reaction is involved: **(Set- 31/2/1)**

(a) Black and White photography (b) Burning of coal

(c) Burning of methane (d) Digestion of food

Q.35. Identify the correct statement about the following reaction: **(Set- 31/2/1)**

$2H_2S + SO_2 \rightarrow 2H_2O + S$

(a) $H_2S$ is oxidising agent and $SO_2$, is reducing agent.

(b) $H_2S$ is reduced to sulphur.

(c) $SO_2$ is oxidising agent and $H_2S$ is reducing agent.

(d) $SO_2$ is oxidised to sulphur.

Q.36. Consider the following Chemical equation: **(Set- 31/3/1)**

a $Al_2O_3$ + b HCl→ c $AlCl_3$ + d $H_2O$

In order to balance this chemical equation, the values of a, b, c and d mustbe

(a) 1, 6, 2 and 3 (b) 1, 6, 3 and 2

(c) 2, 6, 2 and 3 (d) 2, 6, 3 and 2

**Q.37.** Which one of the following reactions is different from the remaining three? **(Set- 31/3/1)**

(a) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$

(b) $CaO + H_2O \rightarrow Ca(OH)_2$

(c) $KNO_3 + H_2SO_4 \rightarrow K_2SO_4 + HNO_3$

(d) $ZnCl_2 + H_2S \rightarrow ZnS + 2HCl$

Q.38. $Zn + 2CH_3COOH \rightarrow (CH_3COO)_2 Zn + H_2$ **(Set- 31/4/1)**

The above reaction is a :

(a) Decomposition reaction (b) Displacement reaction

(c) Double displacement reaction (d) Combination reaction

Q.39. To balance the following chemical equation, the values of the coefficients x, y and z must be respectively: **(Set- 31/5/1)**

$$x\,Zn(NO_3)_2 \xrightarrow{\Delta} y\,ZnO + z\,NO_2 + O_2$$

(a) 4, 2, 2 (b) 4, 4, 2 (c) 2, 2, 4 (d) 2, 4, 2

Q.40. Which of the following is a redox reaction, but not a combination reaction ? **(Set- 31/5/1)**

(a) $C + O_2 \rightarrow CO_2$ (b) $2H_2 + O_2 \rightarrow 2H_2O$

(c) $2Mg + O_2 \rightarrow 2MgO$ (d) $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

**Section-B (two Marker Questions)**

Q.41. Name the type of chemical reaction in which calcium oxide reacts with water. Justify your answer by giving balanced chemical equation for the chemical reaction. **(Set- 31/1/1)**

Hunny Gupta 9999903256

Q.42. (a) No precipitation reaction can occur without exchange of ions between two reactants." Justify this statement giving a balanced chemical equation for the reaction.

**OR** **(Set- 31/2/1)**

(b) Giving one example off each, differentiate between a displacement reaction and a double displacement reaction.

Q.43. When magnesium ribbon is burnt in air, an ash of white colour is produced. Write chemical equation for the reaction giving the chemical name of the ash produced. State the type of chemical reaction giving justification for your answer. **(Set- 31/3/1)**

(a) Copper powder is taken in a china dish and heated over a burner. Name the product formed and state its colour. Write the chemical equation for the reaction involved. **(Set- 31/5/1)**

(b) Write chemical equation for the chemical reaction which occurs when the aqueous solutions of barium chloride and sodium sulphate react together. Write the symbols of the ions present in the compound precipitated in the reaction.

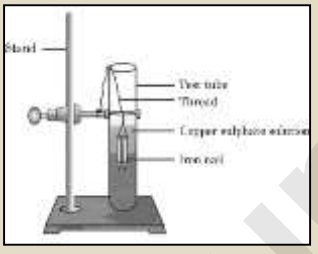
**Section-C (three Marker Questions)**

Q.44. Write one chemical equation each for the chemical reaction in which the following have taken place: **(Set- 31/1/1)**

(i) Change in colour (ii) Change in temperature (iii) Formation of precipitate

Mention colour change/temperature change (rise/fall)/compound precipitated along with equation.

Q.45.




Study the experimental set-up shown in the diagram and write chemical equation for the chemical reaction involved. Name and define the type of reaction. List two other metals which can be used in place of iron to show the same type of reaction with copper sulphate solution. **(Set- 31/5/1)**

**Section-D (five Marker Questions)**

Q.46. (a) What is a chemical reaction? Describe one activity each to show that a Chemical change has occurred in which (i) change of colour, and (ii) change in temperature has taken place. **(Set- 31/5/1)**

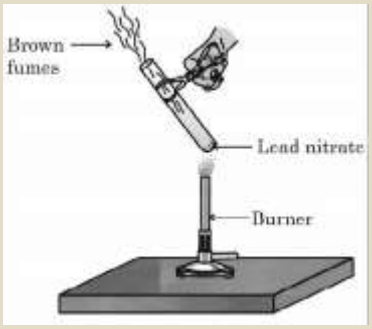
(b) (i) Define a decomposition reaction. How can we say that (I) electrolysis of water, and (II) blackening of silver bromide when exposed to sunlight, are decomposition reactions ? Mention the type of energy involved in each case.

(ii) The type of reactions in which (I) calcium oxide is formed, and (II) calcium hydroxide is formed are opposite reactions to each other. Justify this statement with the help of chemical equations.

Hunny Gupta 9999903256

**Section-A (one Marker Questions)**

Q.47. The emission of brown fumes in the given experimental set-up is due to: **(Set- 31/1/1)**




(a) Thermal decomposition of lead nitrate which produces brown fumes of nitrogen dioxide.
(b) Thermal decomposition of lead nitrate which produces brown fumes of lead oxide.
(c) Oxidation of lead nitrate forming lead oxide and nitrogen dioxide.
(d) Oxidation of lead nitrate forming lead oxide and oxygen.

Q.48. $MnO_2$ +x HC*l* → $MnCl_2$ + y $H_2O$+z $Cl_2$ **(Set- 31/1/1)**
In order to balance the above chemical equation, the values of x, y and z respectively are:
(a) 6, 2, 2 (b) 4, 1, 2 (c) 4, 2, 1 (d) 2, 2, 1

Q.49. **Assertion (A):** The colour of aqueous solution of copper sulphate turns colourless when a Piece of lead is added to it. **(Set- 31/1/1)**
**Reason (R):** Lead is more reactive than copper, and hence displaces copper from its salt solution

Q.50. Which of the following is an example of endothermic process? **(Set- 31/2/1)**
(a) Formation of slaked lime
(b) Decomposition of vegetable matter into compost
(c) Dissolution of ammonium chloride in water
(d) Digestion of food in our body

Q.51. When aqueous solutions of potassium iodide and lead nitrate are mixed, an insoluble substance separates out. The chemical equation for the reaction involved is: **(Set- 31/4/1)**
(a) KI+$PbNO_3$ → PbI+ $KNO_3$ (b) 2KI+ $Pb(NO_3)_2$ → $PbI_2$+ $2KNO_3$
(c) KI+$Pb(NO_3)_2$ → PbI + $KNO_3$ (d) KI+$PbNO_3$ → $PbI_2$+ $KNO_3$

**Q.52.** A metal ribbon 'X' burns in oxygen with a dazzling white flame forming a white ash 'Y'. The correct description of X, Y and the type of reaction is: **(Set- 31/4/1)**
(a) X = Ca; Y = CaO ; Type of reaction = Decomposition
(b) X=Mg; Y = MgO ; Type of reaction = Combination
(c) X= Al; Y = $Al_2O_3$; Type of reaction = Thermal decomposition
(d) X=Zn; Y = ZnO ; Type of reaction = Endothermic

Q.53. The balanced chemical equation showing reaction between quicklime and water is :
(a) 2 CaO + $H_2O$ → 2 CaOH + $H_2$ + Heat
(b) CaO + $H_2O$ → $Ca(OH)_2$ + $H_2$ + Heat
(c) CaO + $H_2O$ → $Ca(OH)_2$ + Heat
(d) 2 CaO + 3 $H_2O$ → 2 $Ca(OH)_3$ + $O_2$ + Heat **(Set- 31/5/1)**

Q.54. Assertion (A): In the following reaction ZnO + C→ Zn + CO , ZnO undergoes reduction.
Reason (R): Carbon is a reducing agent that reduces ZnO to Zn. **(Set- 31/5/1)**

Q.55. Select the appropriate state symbols of the products given as X and Y in the following chemical equation by choosing the correct option from table given below:
$Zn_{(s)}$ + $H_2SO_{4(l)}$ -$ZnSO_{4(X)}$ +$H_{2(Y)}$ **(Set- 31/6/1)**

| | (X) | (Y) |
|---|---|---|
| (a) | (s) | (*l*) |
| (b) | (aq) | (g) |
| (c) | (aq) | (s) |
| (d) | (g) | (aq) |

Q.56. Consider the following chemical equation I and II **(Set- 31/6/1)**

I. $Mg + 2HCl \rightarrow MgCl_2 + H_2$ II. $NaOH + HCl \rightarrow\rightarrow NaCl + H_2O$

The correct statement about these equations is -

(a) T is a displacement reaction and 'II' is a decomposition reaction.

(b) T is a displacement reaction and 'II' is double displacement reaction.

(c) Both 'T' and 'II' are displacement reactions.

(d) Both 'T' and 'II' are double-displacement reactions.

**Section-B (two Marker Questions)**

Q.57. (a) What is observed when aqueous solutions of potassium iodide and lead nitrate are mixed together? Name the type of reaction and write the chemical equation for the reaction that occurs. **(Set- 31/2/1)**

**OR**

(b) When copper powder is heated in a watch glass, a black substance is formed.

(i) Why is this black substance formed? Name it.

(ii) How can this black substance be reversed to its original form

Q.58. State whether the given chemical reaction is a redox reaction or not. Justify your answer.

$MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$ **(Set- 31/5/1)**

**Section-C (three Marker Questions)**

Q.59. Identify the reducing agent in the following reactions: **(Set- 31/5/1)**

(i) $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$ (ii) $H_2O + F_2 \rightarrow HF + HOF$

(iii) $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$ (iv) $2H_2 + O_2 \rightarrow 2H_2O$

(b) Define a redox reaction in terms of gain or loss of oxygen.

Q.60. Silver chloride kept in a china dish turns grey in sunlight. **(Set- 31/2/1)**

(a) Write the colour of silver chloride when it was kept in the china dish.

(b) Name the type of chemical reaction taking place and write the chemical equation for the reaction.

(c) State one use of the reaction. Name one more chemical which can be used for the same purpose.

Q.61. (i) While electrolysing water before passing the current some drops of an acid are added. Why? Name the gases liberated at cathode and anode. Write the relationship between the volume of gas collected at anode and the volume of gas collected at cathode.

(ii) What is observed when silver chloride is exposed to sunlight? Give the type of reaction involved. **(Set- 31/4/1)**

Q.62. (a) Define a double displacement reaction. **(Set- 31/5/1)**

(b) Write the chemical equation of a double displacement reaction which is also a (i) Neutralization reaction and (ii) Precipitation reaction. Give justification for your ans.

**Q.63.** With the help of an appropriate example, justify that some of the chemical reactions are determined by **(Set- 31/6/1)**

(a) Change in temperature, (b) Evolution of a gas, and (c) Change in colour

Give chemical equation for the reaction involved in each case.

Hunny Gupta 9999903256